WORKED SOLUTION

STEP II 2002, Question 6

Pure • Euclidean geometry

STEP II 2002, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/02-s2-q6

The key idea

Everything rests on one fact about an inclined plane: if a line in the plane makes angle ψ with a line of greatest slope, then its angle γ with the horizontal satisfies

$$\sin\gamma = \cos\psi \, \sin\beta,$$

where β is the inclination of the line of greatest slope. Once you have this, each line in the question gives one trigonometric equation, and the perpendicularity of l_1 and l_3 turns a cosine into a sine. After that it is just algebra.

Solution

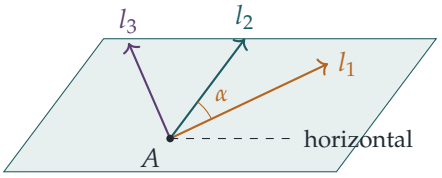
The slope of a line in an inclined plane

Set up axes at A with the z-axis vertical. Let $\mathbf{h}$ be a horizontal unit vector lying in P, and let $\mathbf{u}$ be the unit vector up the line of greatest slope l_2, which is perpendicular to $\mathbf{h}$ in P. Since l_2 makes angle β with the horizontal, we may take

$$\mathbf{h} = (1, 0, 0), \qquad \mathbf{u} = (0, \cos\beta, \sin\beta).$$

A unit vector in P making angle ψ with $\mathbf{u}$ is $\mathbf{d} = \cos\psi\,\mathbf{u} + \sin\psi\,\mathbf{h}$, because $\mathbf{u}$ and $\mathbf{h}$ are orthonormal and span P. Its vertical component is $\cos\psi\sin\beta$. The angle γ between a line with unit direction $\mathbf{d}$ and the horizontal has $\sin\gamma = |\text{vertical component of } \mathbf{d}|$, so

$$\sin\gamma = |\cos\psi| \, \sin\beta. \qquad (*)$$



The first configuration

In P, line l_1 makes acute angle α with l_2, and its angle with the horizontal is $\pi/6$. By $(*)$,

$$\cos\alpha \sin\beta = \sin\tfrac{\pi}{6} = \tfrac{1}{2},$$

as required (here $\cos\alpha > 0$ since α is acute).

Ask yourself What angle does l_3 make with l_2?

All three lines lie in the same plane P and pass through A. Since $l_3 \perp l_1$, the direction of l_3 is that of l_1 turned through a right angle within P, so l_3 makes angle $\alpha \pm \frac{\pi}{2}$ with l_2 (the sign depends on which way we turn, and doesn't matter). Either way $|\cos(\alpha \pm \frac{\pi}{2})| = \sin\alpha$, since α is acute. So $(*)$ for l_3 gives

$$\sin\alpha \sin\beta = \sin\tfrac{\pi}{4} = \tfrac{1}{\sqrt{2}}.$$

Squaring and adding the two results,

$$\sin^2\beta\,(\cos^2\alpha + \sin^2\alpha) = \tfrac{1}{4} + \tfrac{1}{2}, \qquad \text{so} \qquad \sin^2\beta = \tfrac{3}{4}.$$

The angle β between a line and the horizontal lies in $[0, \frac{\pi}{2}]$, so $\sin\beta = \frac{\sqrt{3}}{2}$ and $\beta = \pi/3$, as required. (For the record, dividing the two equations gives $\tan\alpha = \sqrt{2}$, which is indeed acute, so the configuration exists.)

The rotated configuration

The plane, and hence l_2 and $\beta = \pi/3$, are unchanged. Now l_1 makes acute angle θ with l_2 and l_3 makes angle $\theta \pm \frac{\pi}{2}$, exactly as before. Applying $(*)$ to each line, with $\sin\beta = \frac{\sqrt{3}}{2}$:

$$\sin\phi = \tfrac{\sqrt{3}}{2}\cos\theta,$$
$$\sin 2\phi = \tfrac{\sqrt{3}}{2}\sin\theta.$$

Since 2ϕ is an angle with the horizontal, $0 \le 2\phi \le \frac{\pi}{2}$, and $\phi \neq 0$ because $\cos\theta > 0$. So $\sin\phi > 0$ and we may divide the second equation by the first:

$$\tan\theta = \frac{\sin 2\phi}{\sin\phi} = \frac{2\sin\phi\cos\phi}{\sin\phi} = 2\cos\phi.$$

Write $t = \tan^2\theta$, so $\cos^2\phi = t/4$ and $\sin^2\phi = 1 - t/4$. To eliminate θ we use $\cos^2\theta = 1/(1+\tan^2\theta)$ in the square of the first equation:

$$1 - \frac{t}{4} = \frac{3}{4}\cdot\frac{1}{1+t} \quad\Longrightarrow\quad (4-t)(1+t) = 3 \quad\Longrightarrow\quad t^2 - 3t - 1 = 0.$$

Hence $t = \frac{1}{2}\left(3 \pm \sqrt{13}\right)$. Since $t = \tan^2\theta \ge 0$ and $\sqrt{13} > 3$, we must take the plus sign:

$$\tan^2\theta = \frac{3+\sqrt{13}}{2},$$

as required.

The root is genuinely attainable: it gives $\cos^2\phi = t/4 \approx 0.826$, so $\phi \approx 24.7^\circ$, $2\phi \approx 49.4^\circ < 90^\circ$, and $\theta \approx 61.2^\circ$ is acute.

Checks

- In the first configuration, $\tan\alpha = \sqrt{2}$ gives $\cos\alpha = 1/\sqrt{3}$, $\sin\alpha = \sqrt{2/3}$, and with $\sin\beta = \sqrt{3}/2$ the two products are $\frac{1}{2}$ and $1/\sqrt{2}$, as they should be.
- Numerically, $\tan^2\theta \approx 3.3028$, so $\tan\theta \approx 1.8174$, $\theta \approx 61.18^\circ$; then $\sin\phi = \frac{\sqrt{3}}{2}\cos\theta \approx 0.4175$ gives $\phi \approx 24.68^\circ$, and $\frac{\sqrt{3}}{2}\sin\theta \approx 0.7589 \approx \sin(49.35^\circ) = \sin 2\phi$.
- Limiting cases of $(*)$: $\psi = 0$ gives l_2 itself (angle β), and $\psi = \frac{\pi}{2}$ gives a horizontal line (angle 0).

Common slips

- Using $\tan\gamma = \cos\psi\tan\beta$ instead of $\sin\gamma = \cos\psi\sin\beta$. The tangent version is right only if ψ is measured in the horizontal plan view, but here α and θ are angles measured within P.
- Forgetting to reject the negative root $\frac{1}{2}(3 - \sqrt{13})$: a square can't be negative.

Beyond STEP

The formula $(*)$ is really a dot product: the sine of a line's angle with the horizontal is the component of its unit vector along the vertical, and writing that vector in an orthonormal basis of the plane does all the work. Thinking of planes as spans of orthonormal vectors, and of angles as dot products, is the starting point of first-year linear algebra and of the geometry of rotations. Strang's *Introduction to Linear Algebra* develops these ideas carefully.

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